Finite Element Method

Lecture: Kinematics and Strain Measures
MECH4103 — Spring 2026

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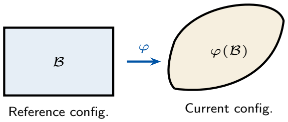
Outline

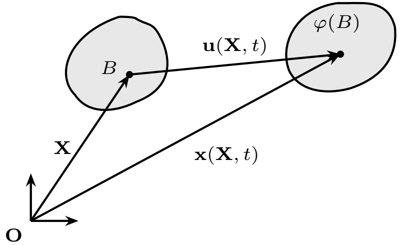
- Classical (linear) FEM assumes **small displacements and small strains**: strains are computed from the linearised strain tensor $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathsf{T}})$.
- Many engineering materials violate this assumption:
  - **Rubber and soft tissues** — strains of 100%–700%
  - **Metal forming, sheet stamping** — large plastic deformation
  - **Flexible structures** — large rotations with small strains
- A **geometrically exact** description of deformation requires new kinematic quantities that reduce to the classical theory only for small displacements.
- These lectures focus on **hyperelastic materials** where no time integration of rate equations is needed — an ideal entry point.



Goal of this lecture

Derive the **deformation gradient** $\mathbf{F}$, strain tensors $\mathbf{E}$ and $\mathbf{C}$, and their geometric meaning.

- A body $\mathcal{B}$ consists of material points $X \in \mathcal{B}$.
- The **reference (initial) configuration**: particle X occupies position $\mathbf{X} = X_A \mathbf{E}_A$ at time $t = 0$.
- The **current (spatial) configuration**: at time t particle X occupies
$$\mathbf{x} = \varphi(\mathbf{X}, t), \qquad x_i = \varphi_i(X_A, t).$$
- The **displacement vector** connects both positions:
$$\mathbf{u}(\mathbf{X}, t) = \varphi(\mathbf{X}, t) - \mathbf{X} = \mathbf{x} - \mathbf{X}.$$
- Capital indices (A, B, C) refer to the **reference** basis $\mathbf{E}_A$; lower indices (i, j, k) refer to the **current** basis $\mathbf{e}_i$.
- **Lagrangian description**: quantities expressed in $\mathbf{X}$ (follow the material).
- **Eulerian description**: quantities expressed in $\mathbf{x}$ (fixed spatial point).



Definition

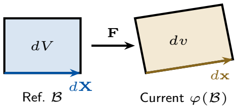
The **deformation gradient** $\mathbf{F}$ maps an infinitesimal material line element $d\mathbf{X}$ in $\mathcal{B}$ to the corresponding element $d\mathbf{x}$ in $\varphi(\mathcal{B})$:

$$\boxed{d\mathbf{x} = \mathbf{F}\, d\mathbf{X}}, \qquad F_{iA} = \frac{\partial x_i}{\partial X_A}$$

- $\mathbf{F} = \mathrm{Grad}\, \varphi(\mathbf{X}, t) = \mathbf{1} + \mathrm{Grad}\, \mathbf{u}$, where $\mathrm{Grad}(\cdot)$ is the gradient with respect to $\mathbf{X}$.
- Matrix form:

$$[F_{iA}] = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \\ x_{3,1} & x_{3,2} & x_{3,3} \end{bmatrix}$$

- The **Jacobian determinant** $J = \det \mathbf{F} > 0$ ensures non-penetration and invertibility.
- Volume transformation: $dv = J\, dV$.
- Nanson's formula (surface normal transformation): $d\mathbf{a} = J\mathbf{F}^{-\mathsf{T}}\, d\mathbf{A}$.



Physical meaning

$\mathbf{F}$ encodes all local information about stretching *and* rotation at a material point.

Right Cauchy–Green tensor

$$\mathbf{C} = \mathbf{F}^{\mathsf{T}}\mathbf{F}, \qquad C_{AB} = F_{iA}F_{iB}$$

- Relates squared lengths: $|d\mathbf{x}|^2 = d\mathbf{X} \cdot \mathbf{C}\, d\mathbf{X}$.
- **Symmetric**, **positive-definite**.
- Referred to the **reference** configuration.

Green–Lagrange strain tensor

$$\boxed{\mathbf{E} = \tfrac{1}{2}(\mathbf{C} - \mathbf{1}) = \tfrac{1}{2}(\mathbf{F}^{\mathsf{T}}\mathbf{F} - \mathbf{1})}$$

- Measures change of squared length from $\mathcal{B}$ to $\varphi(\mathcal{B})$.
- In terms of displacement gradient $\mathbf{H} = \operatorname{Grad}\mathbf{u}$:

$$\mathbf{E} = \tfrac{1}{2}\left(\mathbf{H} + \mathbf{H}^{\mathsf{T}} + \mathbf{H}^{\mathsf{T}}\mathbf{H}\right)$$

- The term $\mathbf{H}^{\mathsf{T}}\mathbf{H}$ is the *nonlinear* (geometric) part.
- For **small** $\|\mathbf{H}\| \ll 1$: $\mathbf{E} \to \boldsymbol{\varepsilon} = \tfrac{1}{2}(\mathbf{H} + \mathbf{H}^{\mathsf{T}})$ (linear theory).

Component form

$$E_{AB} = \tfrac{1}{2}(F_{iA}F_{iB} - \delta_{AB})$$

Key property: rigid-body invariance

For a pure rigid-body motion $\varphi_R = \mathbf{Q}\mathbf{X} + \mathbf{c}$ $(\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{1})$:

$$\mathbf{F}_R = \mathbf{Q} \;\Rightarrow\; \mathbf{C}_R = \mathbf{1} \;\Rightarrow\; \mathbf{E}_R = \mathbf{0}.$$

The Green–Lagrange strain is identically **zero** for any rigid-body motion — the linear strain $\boldsymbol{\varepsilon}$ is not.

The left Cauchy–Green tensor $\mathbf{b} = \mathbf{F}\mathbf{F}^{\mathsf{T}}$ plays the analogous role referred to the **current** configuration.

Polar decomposition

$\mathbf{F}$ can be split *uniquely* into a proper orthogonal rotation $\mathbf{R}$ and a symmetric positive-definite stretch tensor:

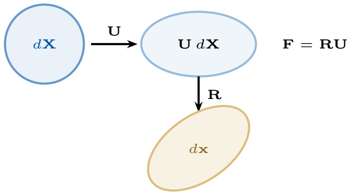
$$\mathbf{F} = \mathbf{R}\,\mathbf{U} = \mathbf{V}\,\mathbf{R}$$

- $\mathbf{U}$ — right stretch tensor (reference config.)
- $\mathbf{V}$ — left stretch tensor (current config.)
- $\mathbf{C} = \mathbf{U}^2, \quad \mathbf{b} = \mathbf{V}^2$

Spectral decomposition

$$\mathbf{U} = \sum_{i=1}^{3} \lambda_i\,\mathbf{N}_i \otimes \mathbf{N}_i, \qquad \mathbf{V} = \sum_{i=1}^{3} \lambda_i\,\mathbf{n}_i \otimes \mathbf{n}_i$$

- $\lambda_i > 0$ — **principal stretches**
- $\mathbf{N}_i$ — principal directions in $\mathcal{B}$
- $\mathbf{n}_i = \mathbf{R}\mathbf{N}_i$ — principal directions in $\varphi(\mathcal{B})$
- $\mathbf{C} = \sum_i \lambda_i^2\,\mathbf{N}_i \otimes \mathbf{N}_i$



Interpretation

Any deformation = pure stretch ($\mathbf{U}$) followed by pure rotation ($\mathbf{R}$). Only the stretch contributes to strain.

- Green–Lagrange ($\mathbf{E}$) and Almansi ($\mathbf{e}$) are special cases of a one-parameter family:

$$\mathbf{E}^{(\alpha)} = \frac{1}{\alpha}\left(\mathbf{U}^{\alpha} - \mathbf{1}\right), \quad \alpha \in \mathbb{R} \qquad \longleftrightarrow \qquad \mathbf{e}^{(\alpha)} = \frac{1}{\alpha}\left(\mathbf{V}^{\alpha} - \mathbf{1}\right)$$

- In terms of principal stretches via spectral decomposition:

$$\mathbf{E}^{(\alpha)} = \frac{1}{\alpha}\sum_{i=1}^{3}(\lambda_i^{\alpha} - 1)\,\mathbf{N}_i \otimes \mathbf{N}_i$$

α	Strain measure	Name
$+2$	$\mathbf{E} = \frac{1}{2}(\mathbf{U}^2 - \mathbf{1})$	Green–Lagrange
-2	$\mathbf{e} = \frac{1}{2}(\mathbf{1} - \mathbf{V}^{-2})$	Almansi
$+1$	$\mathbf{U} - \mathbf{1}$	Biot
$\to 0$	$\ln \mathbf{U}$	Hencky (logarithmic)

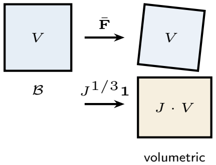
- All measures coincide for small strains: $\mathbf{E}^{(\alpha)} \to \boldsymbol{\varepsilon}$ as $\|\mathbf{H}\| \to 0$.
- Hencky strain ($\alpha \to 0$) is particularly natural for **finite-strain plasticity** and multiplicative decomposition.
- **For hyperelastic models** (these lectures), Green–Lagrange $\mathbf{E}$ and right Cauchy–Green $\mathbf{C}$ are the preferred measures.

Multiplicative split (Flory 1961)

$$\mathbf{F} = \underbrace{J^{1/3}\mathbf{1}}_{\text{volume change}} \cdot \underbrace{\bar{\mathbf{F}}}_{\text{shape change}}, \quad \bar{\mathbf{F}} = J^{-1/3}\mathbf{F}, \quad \det \bar{\mathbf{F}} = 1$$

Corresponding split of $\mathbf{C}$:

$$\bar{\mathbf{C}} = \bar{\mathbf{F}}^{\mathsf{T}}\bar{\mathbf{F}} = J^{-2/3}\mathbf{C}$$



- Rubber and biological tissues are **nearly incompressible**: $J \approx 1$.
- Incompressibility constraint: $J = 1 \Leftrightarrow \det \mathbf{F} = 1$.
- The split decouples bulk and shear responses in the strain energy — crucial for **mixed FEM** formulations (next lecture).
- Corresponds to the additive split in the linear theory: $\boldsymbol{\varepsilon} = \mathbf{e}^D + \frac{1}{3}(\operatorname{tr}\boldsymbol{\varepsilon})\mathbf{1}$.

Why it matters in FEM

Nearly incompressible behaviour causes **volumetric locking** in standard displacement elements — the split guides the remedy.

Exercise 1 — Plane Deformation: Compute $\mathbf{F}$ and $\mathbf{E}$

Exercise

A body undergoes a *plane deformation* described by

$$x_1 = X_1 + u_1(X_1, X_2), \qquad x_2 = X_2 + u_2(X_1, X_2), \qquad x_3 = X_3.$$

(a) Determine the matrix form of the deformation gradient $\mathbf{F}$.
(b) Compute the components of the Green–Lagrange strain tensor $\mathbf{E}$.

Solution

(a) From $F_{iA} = \partial x_i / \partial X_A$:

$$[\mathbf{F}] = \begin{bmatrix} 1 + u_{1,1} & u_{1,2} & 0 \\ u_{2,1} & 1 + u_{2,2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) $\mathbf{E} = \frac{1}{2}(\mathbf{F}^\mathsf{T}\mathbf{F} - \mathbf{1})$. Computing $\mathbf{F}^\mathsf{T}\mathbf{F}$ and subtracting $\mathbf{1}$:

$$E_{11} = u_{1,1} + \tfrac{1}{2}(u_{1,1}^2 + u_{2,1}^2)$$

$$E_{22} = u_{2,2} + \tfrac{1}{2}(u_{2,2}^2 + u_{1,2}^2)$$

$$E_{12} = \tfrac{1}{2}(u_{1,2} + u_{2,1}) + \tfrac{1}{2}(u_{1,1}u_{1,2} + u_{2,2}u_{2,1})$$

The *first* terms reproduce linear strain theory; the quadratic terms capture finite-deformation effects. For small $\|\operatorname{Grad}\mathbf{u}\|$, all quadratic terms vanish.

Spatial velocity gradient $\mathbf{l}$

$$\mathbf{l} = \dot{\mathbf{F}}\mathbf{F}^{-1} = \operatorname{grad}\mathbf{v}, \qquad l_{ij} = \frac{\partial v_i}{\partial x_j}$$

Additive split into symmetric and skew parts:

$$\mathbf{l} = \underbrace{\mathbf{d}}_{\text{rate of deformation}} + \underbrace{\mathbf{w}}_{\text{spin (rotation rate)}}$$

$$\mathbf{d} = \tfrac{1}{2}(\mathbf{l} + \mathbf{l}^{\mathsf{T}}), \quad \mathbf{w} = \tfrac{1}{2}(\mathbf{l} - \mathbf{l}^{\mathsf{T}})$$

Rate of Green–Lagrange strain

$$\dot{\mathbf{E}} = \mathbf{F}^{\mathsf{T}}\,\mathbf{d}\,\mathbf{F}$$

$\mathbf{d}$ pulled back to the **fixed** reference config. gives $\dot{\mathbf{E}}$: the work-conjugate rate to $\mathbf{S}$.

Do we need rates for static hyperelasticity?

- **No** rate constitutive law: stress follows directly from $\mathbf{S} = 2\,\partial W/\partial\mathbf{C}$ — no time integration.
- **But** $\mathbf{d}$ enters the *linearisation* of the weak form (tangent stiffness, Lecture 2).
- For **dynamics** ($\rho_0\,\ddot{\mathbf{u}} \neq \mathbf{0}$) the full velocity field is needed.
- The spin $\mathbf{w}$ is essential for *objective stress rates* in elasto-plasticity — introduced next.

Summary: what we carry forward

Quantity	Used for
$\mathbf{d}$	tangent stiffness, dynamics
$\dot{\mathbf{E}} = \mathbf{F}^{\mathsf{T}}\mathbf{d}\mathbf{F}$	work conjugacy with $\mathbf{S}$
$\mathbf{w}$	objective stress rates (plasticity)

What is a superposed rigid-body motion?

A second observer views the *same* body through a time-dependent rotation $\mathbf{Q}(t)$ and translation $\mathbf{c}(t)$:

$$\mathbf{x}^* = \mathbf{Q}(t)\,\mathbf{x} + \mathbf{c}(t), \quad \mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{1},\ \det\mathbf{Q} = +1$$

- $\mathbf{Q}(t)$: **rotation matrix varying in time**
  Think: a camera spinning around the body.
- $\mathbf{c}(t)$: **translation vector varying in time**
  Think: the camera also moving sideways.

Objectivity — consistent mapping between frames

A quantity is **objective** if it can be mapped consistently between coordinate systems using $\mathbf{Q}(t)$, so all observers describe the *same physical behaviour*:

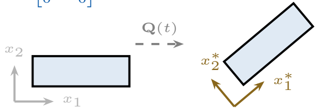
- Objective **vector**: $\mathbf{a}^* = \mathbf{Q}\mathbf{a}$
- Objective **tensor**: $\boldsymbol{\sigma}^* = \mathbf{Q}\boldsymbol{\sigma}\mathbf{Q}^\mathsf{T}$
- $\mathbf{S}$, $\mathbf{E}$, $\mathbf{C}$ always objective: live in the fixed reference config., never see $\dot{\mathbf{Q}}$.

Observer 1 (fixed)

$\boldsymbol{\sigma}_0 = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$, $\dot{\boldsymbol{\sigma}} = \mathbf{0}$

Observer 2 (spinning ω)

$\boldsymbol{\sigma}^* = \mathbf{Q}\boldsymbol{\sigma}_0\mathbf{Q}^\mathsf{T}$ ✓ objective



Problem: compute the stress rate

Plain rate $\dot{\boldsymbol{\sigma}}$ × **NOT objective**

Observer 1: $\dot{\boldsymbol{\sigma}} = \mathbf{0}$ Observer 2: $\dot{\boldsymbol{\sigma}}^* \neq \mathbf{0}$

Why? — see next slide

Why $\dot{\boldsymbol{\sigma}}$ Is Not Objective — A Concrete Example

Setup: block at rest, Observer 2 spins at rate ω

$$\boldsymbol{\sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad \text{(constant, } \dot{\boldsymbol{\sigma}} = \mathbf{0}\text{)}$$

Observer 2 rotates: $\mathbf{Q}(t) = \begin{bmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{bmatrix}$

Observer 2 measures (correctly):

$$\boldsymbol{\sigma}^*(t) = \mathbf{Q}\boldsymbol{\sigma}\mathbf{Q}^\mathsf{T} = \begin{bmatrix} \cos^2\omega t & \sin\omega t\cos\omega t \\ \sin\omega t\cos\omega t & \sin^2\omega t \end{bmatrix}$$

Components oscillate — **this is correct**: same physical state, different coordinate description. Objectivity holds for $\boldsymbol{\sigma}$.

Observer 2 computes $\dot{\boldsymbol{\sigma}}^*$ (at $t = 0$):

Differentiating $\boldsymbol{\sigma}^* = \mathbf{Q}\boldsymbol{\sigma}\mathbf{Q}^\mathsf{T}$:

$$\dot{\boldsymbol{\sigma}}^* = \underbrace{\mathbf{Q}\dot{\boldsymbol{\sigma}}\mathbf{Q}^\mathsf{T}}_{=\mathbf{0}} + \underbrace{\dot{\mathbf{Q}}\boldsymbol{\sigma}\mathbf{Q}^\mathsf{T} + \mathbf{Q}\boldsymbol{\sigma}\dot{\mathbf{Q}}^\mathsf{T}}_{\text{spurious spin terms}}$$

At $t = 0$ ($\mathbf{Q} = \mathbf{1}$, $\dot{\mathbf{Q}} = \omega \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$):

$$\dot{\boldsymbol{\sigma}}^*\big|_{t=0} = \omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \neq \mathbf{0}$$

The failure

Observer 1: $\dot{\boldsymbol{\sigma}} = \mathbf{0}$ (nothing happening).
Observer 2: $\dot{\boldsymbol{\sigma}}^* \neq \mathbf{0}$ (purely from spinning).

Two observers disagree about the stress rate even though the **material is doing nothing**. The extra ω term is a kinematic artifact of the rotating frame — not physical.

A constitutive law $\dot{\boldsymbol{\sigma}} = \ : \mathbf{d}$ would predict different material behaviour depending on how fast the observer spins. **Physically absurd.**

The Lie Derivative — Pull Back, Differentiate, Push Forward

The key insight

The current configuration **rotates with the body**. Differentiating a spatial tensor there introduces $\dot{\mathbf{Q}}$ — the observer-spin contamination.

Solution: step out of the rotating frame entirely.

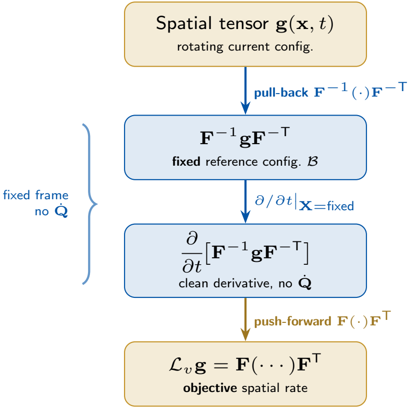
Three-step recipe (Lie derivative)

$$\mathcal{L}_v \mathbf{g} := \mathbf{F}\left(\frac{\partial}{\partial t}\Big[\mathbf{F}^{-1}\mathbf{g}\mathbf{F}^{-\mathsf{T}}\Big]\right)\mathbf{F}^{\mathsf{T}}$$

1. **Pull back** $\mathbf{F}^{-1}(\cdot)\mathbf{F}^{-\mathsf{T}}$: leave the rotating current frame, enter the **fixed** reference frame.
2. **Differentiate** $\partial/\partial t|_{\mathbf{X}}$: in the fixed frame $\dot{\mathbf{Q}} = \mathbf{0}$ — no contamination.
3. **Push forward** $\mathbf{F}(\cdot)\mathbf{F}^{\mathsf{T}}$: bring the clean result back to the current frame.

Merry-go-round analogy

Measuring velocity while spinning gives contaminated results. Step off, measure on solid ground, translate back. That translation back is the push-forward.



Result: $\mathcal{L}_v \mathbf{e} = \mathbf{d}$ (Almansi rate = rate of deformation)

Same setup as before

$$\boldsymbol{\sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \dot{\boldsymbol{\sigma}} = \mathbf{0}, \quad \mathbf{l} = \mathbf{d} = \mathbf{w} = \mathbf{0}$$

The body is at rest: no deformation, no velocity gradient.

For a rigid-body motion $\varphi_R = \mathbf{Q}(t)\mathbf{x} + \mathbf{c}(t)$ the deformation gradient is $\mathbf{F} = \mathbf{Q}(t)$, so

$$\mathbf{F}^{-1} = \mathbf{Q}^{\mathsf{T}}, \quad \dot{\mathbf{F}} = \dot{\mathbf{Q}}, \quad \mathbf{l} = \dot{\mathbf{F}}\mathbf{F}^{-1} = \dot{\mathbf{Q}}\mathbf{Q}^{\mathsf{T}}.$$

Step 1 — Pull back $\boldsymbol{\sigma}$ to reference config.

$$\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-\mathsf{T}} = \mathbf{Q}^{\mathsf{T}}\boldsymbol{\sigma}\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{(constant!)}$$

Because $\boldsymbol{\sigma}^* = \mathbf{Q}\boldsymbol{\sigma}\mathbf{Q}^{\mathsf{T}}$, the pull-back simply undoes the rotation and recovers the **original constant tensor**.

Step 2 — Differentiate in the reference frame

$$\frac{\partial}{\partial t}\left[\mathbf{Q}^{\mathsf{T}}\boldsymbol{\sigma}\mathbf{Q}\right] = \mathbf{0}$$

The pulled-back tensor is constant $\Rightarrow$ its time derivative is exactly **zero**. No $\dot{\mathbf{Q}}$ term appears.

Step 3 — Push forward back to current config.

$$\mathcal{L}_v\boldsymbol{\sigma} = \mathbf{F} \cdot \mathbf{0} \cdot \mathbf{F}^{\mathsf{T}} = \mathbf{0}$$

Conclusion

Both observers compute $\mathcal{L}_v\boldsymbol{\sigma} = \mathbf{0}$. The Lie derivative correctly reports **zero rate of stress change** — because the material is doing nothing.

Compare: $\dot{\boldsymbol{\sigma}}^*|_{t=0} = \omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \neq \mathbf{0}$ (the plain rate gave a spurious result).

Starting from the definition

$$\mathcal{L}_v \boldsymbol{\tau} := \mathbf{F}\left(\frac{\partial}{\partial t}\left[\mathbf{F}^{-1}\boldsymbol{\tau}\mathbf{F}^{-\mathsf{T}}\right]\right)\mathbf{F}^{\mathsf{T}}$$

Apply the product rule to the time derivative inside:

$$\frac{\partial}{\partial t}[\mathbf{F}^{-1}\boldsymbol{\tau}\mathbf{F}^{-\mathsf{T}}] = \dot{\mathbf{F}}^{-1}\boldsymbol{\tau}\mathbf{F}^{-\mathsf{T}} + \mathbf{F}^{-1}\dot{\boldsymbol{\tau}}\mathbf{F}^{-\mathsf{T}} + \mathbf{F}^{-1}\boldsymbol{\tau}\dot{\mathbf{F}}^{-\mathsf{T}}$$

Use the identity $\dot{\mathbf{F}}^{-1} = -\mathbf{F}^{-1}\dot{\mathbf{F}}\mathbf{F}^{-1}$

This follows from differentiating $\mathbf{F}^{-1}\mathbf{F} = \mathbf{1}$:

$$\dot{\mathbf{F}}^{-1} = -\mathbf{F}^{-1}\dot{\mathbf{F}}\mathbf{F}^{-1} = -\mathbf{F}^{-1}\mathbf{l} \qquad (\text{since } \mathbf{l} = \dot{\mathbf{F}}\mathbf{F}^{-1})$$

Substituting back and pushing forward by $\mathbf{F}(\cdots)\mathbf{F}^{\mathsf{T}}$:

$$\mathbf{F}\dot{\mathbf{F}}^{-1}\boldsymbol{\tau}\mathbf{F}^{-\mathsf{T}}\mathbf{F}^{\mathsf{T}} = -\mathbf{l}\boldsymbol{\tau}$$

$$\mathbf{F}\mathbf{F}^{-1}\boldsymbol{\tau}\dot{\mathbf{F}}^{-\mathsf{T}}\mathbf{F}^{\mathsf{T}} = -\boldsymbol{\tau}\mathbf{l}^{\mathsf{T}}$$

$$\mathbf{F}\mathbf{F}^{-1}\dot{\boldsymbol{\tau}}\mathbf{F}^{-\mathsf{T}}\mathbf{F}^{\mathsf{T}} = \dot{\boldsymbol{\tau}}$$

Final result

$$\boxed{\mathcal{L}_v \boldsymbol{\tau} = \dot{\boldsymbol{\tau}} - \mathbf{l}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{l}^{\mathsf{T}}}$$

This is the **Oldroyd rate** of the Kirchhoff stress.

Physical interpretation

- $\dot{\boldsymbol{\tau}}$: total (non-objective) time derivative.
- $-\mathbf{l}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{l}^{\mathsf{T}}$: correction terms that subtract the spin contamination introduced by $\mathbf{l} = \mathbf{d} + \mathbf{w}$.
- The correction uses the **full** velocity gradient $\mathbf{l}$, not just the spin $\mathbf{w}$.
- For a **rigid body**: $\mathbf{l} = \dot{\mathbf{Q}}\mathbf{Q}^{\mathsf{T}}$, $\dot{\boldsymbol{\tau}} = \mathbf{l}\boldsymbol{\tau} + \boldsymbol{\tau}\mathbf{l}^{\mathsf{T}}$ (from differentiating $\boldsymbol{\tau}^* = \mathbf{Q}\boldsymbol{\tau}\mathbf{Q}^{\mathsf{T}}$), so $\mathcal{L}_v\boldsymbol{\tau} = \mathbf{0}$ — consistent with the example on the previous slide.

Three objective stress rates

Rate	Closed form
Lie	$\mathcal{L}_v\boldsymbol{\tau} = \dot{\boldsymbol{\tau}} - \mathbf{l}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{l}^{\mathsf{T}}$
Jaumann	$\overset{\nabla}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \mathbf{w}\boldsymbol{\sigma} + \boldsymbol{\sigma}\mathbf{w}$
Green–Naghdi	$\overset{\circ}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \boldsymbol{\Omega}\boldsymbol{\sigma} + \boldsymbol{\sigma}\boldsymbol{\Omega}, \;\; \boldsymbol{\Omega} = \dot{\mathbf{R}}\mathbf{R}^{\mathsf{T}}$

Key difference: which spin is subtracted?

- **Lie**: uses full $\mathbf{l} = \mathbf{d} + \mathbf{w}$ (stretch *and* spin). Pulls back to **fixed** $\mathcal{B}$. Exact.
- **Green–Naghdi**: uses $\boldsymbol{\Omega} = \dot{\mathbf{R}}\mathbf{R}^{\mathsf{T}}$ from polar decomposition. Co-rotates with material rotation. Partial geometric meaning.
- **Jaumann**: uses only spin $\mathbf{w}$. No pull-back — purely algebraic correction.

Relation between Lie and Jaumann

Splitting $\mathbf{l} = \mathbf{d} + \mathbf{w}$:

$$\mathcal{L}_v\boldsymbol{\tau} = \underbrace{\dot{\boldsymbol{\tau}} - \mathbf{w}\boldsymbol{\tau} + \boldsymbol{\tau}\mathbf{w}}_{\overset{\nabla}{\boldsymbol{\tau}}\ \text{(Jaumann)}} - \mathbf{d}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{d}$$

$$\Rightarrow \quad \mathcal{L}_v\boldsymbol{\tau} = \overset{\nabla}{\boldsymbol{\tau}} - \mathbf{d}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{d}$$

They differ by the **stretching terms** $\mathbf{d}\boldsymbol{\tau} + \boldsymbol{\tau}\mathbf{d}$. For **small strains** $(\mathbf{d} \approx \mathbf{0})$ all rates coincide.

Practical guidance

- **This course** (hyperelasticity): Lie rate appears naturally in the tangent stiffness.
- **Metal plasticity**: Jaumann is the default in Abaqus/Explicit and LS-DYNA — simple but has a known pathology: *oscillating shear stress* under large simple shear (Nagtegaal & de Jong 1981).
- **Modern large-strain plasticity**: Lie rate with multiplicative split $\mathbf{F} = \mathbf{F}_e\mathbf{F}_p$ is preferred.

Exercise

A plate loaded in-plane has deformation gradient

$$[\mathbf{F}] = \begin{bmatrix} 3 & -1 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Determine **(a)** the principal stretches λ_i, **(b)** the right stretch tensor $\mathbf{U}$, and **(c)** the rotation tensor $\mathbf{R}$.

Solution

(a) Compute $\mathbf{C} = \mathbf{F}^\mathsf{T}\mathbf{F}$:

$$[\mathbf{C}] = \begin{bmatrix} 13 & 1 & 0 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Eigenvalues of $\mathbf{C}$ satisfy $(1-\lambda^2)(\lambda^4 - 18\lambda^2 + 64) = 0$, giving

$$\lambda_{1,2}^2 = 9 \pm \sqrt{17}, \qquad \lambda_3 = 1.$$

(b) Eigenvectors $\mathbf{N}_i$ of $\mathbf{C}$ are determined from $(\mathbf{C} - \lambda_i^2 \mathbf{1})\mathbf{N}_i = \mathbf{0}$; then $\mathbf{U} = \sum_i \lambda_i \, \mathbf{N}_i \otimes \mathbf{N}_i \;\Rightarrow [\mathbf{U}] \approx \left[\begin{smallmatrix} 3.60 & 0.17 & 0 \\ 0.17 & 2.23 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$.

(c) $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$; for plane deformation $\tan\theta = (F_{12} - F_{21})/(F_{11} + F_{22})$ gives $\theta \approx -31°$.

Key quantities introduced

- **Deformation gradient**: $\mathbf{F} = \mathrm{Grad}\,\varphi$, $d\mathbf{x} = \mathbf{F}\,d\mathbf{X}$
- **Jacobian**: $J = \det \mathbf{F}$, volume ratio $dv = J\,dV$
- **Right Cauchy–Green tensor**: $\mathbf{C} = \mathbf{F}^{\mathsf{T}}\mathbf{F}$
- **Green–Lagrange strain**: $\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{1})$
- **Polar decomposition**: $\mathbf{F} = \mathbf{R}\mathbf{U}$, principal stretches λ_i
- **Vol–iso split**: $\mathbf{F} = J^{1/3}\bar{\mathbf{F}}$, $\bar{\mathbf{C}} = J^{-2/3}\mathbf{C}$

Connections to linear theory

- $\mathbf{E} \to \boldsymbol{\varepsilon}$ for $\|\mathrm{Grad}\,\mathbf{u}\| \ll 1$
- $J \to 1 + \mathrm{Div}\,\mathbf{u}$ for small displacements
- Linear elastic incompressibility: $\mathrm{tr}\,\boldsymbol{\varepsilon} = 0 \;\leftrightarrow\; J = 1$

Preview: Lecture 2

In the next lecture we use these kinematic measures to:

1. Define stress tensors work-conjugate to $\mathbf{E}$ and $\mathbf{C}$ (2nd Piola–Kirchhoff, Cauchy, Kirchhoff)
2. Derive hyperelastic constitutive models: Neo-Hooke, Mooney–Rivlin, Ogden
3. Formulate the **weak form of equilibrium** and its linearisation for FEM

Reference

P. Wriggers, *Nonlinear Finite Element Methods*, Springer 2008, Chapter 3 (Sections 3.1–3.2).

Thanks for your attention!